

TERRY SCHERMERHORN

AI Systems Architect · Multi-Agent & Federated AI · Independent Researcher · U.S. Air Force Veteran
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PROFESSIONAL SUMMARY

Systems-oriented technologist, independent researcher, and U.S. Air Force veteran with over 30 years of operational depth across military weapons systems, electronics, database management, and technical leadership. Currently specializing in sovereign, federated, and verifiable artificial intelligence infrastructure — designing systems that prioritize local control, auditability, and resilience without cloud dependency.

Creator of **FI-13** — an open-source 13-protocol federated interoperability stack for heterogeneous sovereign AI systems — published on Zenodo (DOI: 10.5281/zenodo.20262505) and demonstrated through three working prototypes running entirely on local commodity hardware. Active researcher with six published papers spanning federated AI architecture, cislunar systems, and cosmology.

CORE COMPETENCIES

Federated AI Infrastructure: Protocol design, sovereignty-preserving interoperability, multi-agent orchestration, distributed consensus

Systems Architecture: Failure mode analysis, cascade modeling, root-cause diagnostics, governance primitives, edge-first deployment

Technical Research: LaTeX specification, peer-ready academic publishing, open-science workflows, cryptographic verification primitives

IT Infrastructure: Local edge-compute optimization, Docker, database monitoring, GPU-accelerated environments

Leadership & Communication: Multi-disciplinary team leadership, executive briefing, requirements translation, mission-critical operations

Electronics & Maintenance: ISO 9001, Lean Six Sigma, fine pitch soldering

TECHNICAL STACK

Languages & Environments: Python, JavaScript, Bash, SQL, WSL, Ubuntu Linux, Windows 11

AI Tooling & Compute: Ollama, vLLM, local LLM cluster management, GPU-accelerated inference, FAISS, sentence-transformers

3D Modeling: Lightwave 3d, Blender

Databases & DevOps: SQLite, MySQL, Docker, GitHub version control, Linux system optimization

Research & Documentation: LaTeX, BibTeX, Zenodo, ORCID, academic publishing pipelines

PROFESSIONAL EXPERIENCE

Principal AI Systems Architect & Independent Researcher

2023 – Present

- Built Home lab featuring a Ryzen 9 7950 cpu, RTX 4070, and 64 gb ram
- Originated and led development of **FI-13** — a 13-protocol federated AI interoperability stack establishing decentralized governance primitives, Byzantine fault isolation, and sovereignty-preserving context exchange across heterogeneous AI nodes.
- Built **Methuselah** — a locally-sovereign AI dashboard with autonomous Gatekeeper LLM performing prompt scrubbing, intent classification, model routing, and dynamic weight adjustment entirely on local hardware with no cloud dependencies.
- Developed the **Shackleton Protocol** — a constitution-governed multi-agent think tank with five parallel domain specialists (Engineering, Biology, Geology, Logistics, IT) producing ranked, feasibility-scored solution architectures for complex engineering problems.

- Designed and implemented a multi-model Specialist Debate engine (**Hive Engine**) implementing P10/P11 FI-13 protocols — heterogeneous LLMs running structured ASSERT → COUNTER → VALIDATE cycles with provenance-tracked verdicts.
- Published six open-access research papers on Zenodo spanning federated AI architecture, cislunar systems infrastructure, cryogenic quantum computing, and the Theta-Psi geometric cosmological framework.
- All systems developed and validated on sovereign local hardware: AMD Ryzen 9 7950X / RTX 4070 / 64GB DDR5.

Founder | Paradigm 3D**2007 – 2023**

- Founded and operate an independent technical consultancy delivering 3D modeling, rapid prototyping, visualization, and remote IT support to long-standing clients since 2007.
- Delivered custom 3D modeling services for Dune Studios contributing to significant Q4 2015 revenue increase.

Facilities Technician III | Harrah's Casino, Pompano Beach, FL**2016 – 2023**

- Led full facility integration of IBM Maximo CMMS enterprise asset management system, developing custom asset tracking structures and training technical staff.
- Managed after-hours emergency response for complex electromechanical system failures — maintained consistent record of zero systemic operational downtime.
- Spearheaded facility-wide emergency lighting inspection, labeling, and replacement program across entire property.
- Led emergency fire suppression and personnel evacuation — instrumental in preventing escalation and ensuring personnel safety.

Missile Systems Analyst Supervisor | USAF & Air National Guard**1981 – 2010**

- Maintained and performed nuclear-capable platform diagnostics on AGM-86B Air Launched Cruise Missile, AIM-120B AMRAAM, AIM-9X Sidewinder, and Short-Range Attack Missile (SRAM) under **Top Secret clearance**.
- Performed post-loading checks on B-52G, F-16, and F-15 — ensuring 100% safety and operational readiness on nuclear-capable platforms.
- Discovered and eliminated a **\$500,000 database error** in the Core Automated Maintenance System (CAMS) — preventing significant financial loss and preserving mission readiness.
- NCOIC of Equipment Maintenance section — led and supervised team of six specialists under stringent military compliance guidelines.
- Volunteered for special duty assignment in the immediate aftermath of the September 11, 2001 terror attacks.
- Awarded five Air Force Achievement Medals for leadership, precision diagnostic execution, and operational excellence.

PUBLICATIONS

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- **Federated AI Infrastructure: Layered Interoperability Protocols for Sovereign Machine Intelligence**
Schermerhorn, T. (2026). Zenodo. DOI: 10.5281/zenodo.20262505
 - **The 74/26 Principle: Entropy Minimization and Physics-Constrained Stability in Cislunar Resource Networks**
Schermerhorn, T. (2026). Zenodo. DOI: 10.5281/zenodo.15207657
 - **Cryogenic Infrastructure for Lunar-Scale Quantum Computing; The Theta-Psi Framework Papers I–III**
Full publication record: zenodo.org/search?q=Schermerhorn • ORCID: 0009-0004-1981-0916

EDUCATION & CREDENTIALS

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- **Master of Science in Business & Entrepreneurship** | Full Sail University | 2015
 - **Master of Science in Information Systems** | University of Phoenix | 2011
 - **Bachelor of Science in Information Technology** | University of Phoenix | 2009
 - **Security Clearance:** Top Secret / SBI (Historical Holder)
 - ORCID: 0009-0004-1981-0916
 - [Github.com/Terry2023](https://github.com/Terry2023)
 - [Zenodo.org/records/20262505](https://zenodo.org/records/20262505)